



Fatemeh Hassanvand

Curriculum Vitae, May 2026

University of Udine, Via delle Scienze, 206 - 33100 - Udine, Italy



fatemeh.hassanvand@uniud.it, hassanvand3250@gmail.com



LinkedIn



GitHub

RESEARCH INTERESTS

Machine Learning; Explainable AI; Multi-Agent Systems; Human-AI Interaction; Value-Aligned AI; Agent-Based Modeling; Decision-Making under Uncertainty

EDUCATION

University of Isfahan (UI), Isfahan, Iran

2018 - 2022

M.Sc. in IT Engineering-Electronic Commerce

GPA: A+

Thesis Title: *"Effectively Modeling of Singly-Peaked Opponents in Automated Bilateral Negotiations"*

Supervisor: Dr. Faria Nassiri-Mofakham

Lorestan University (LU), Khorramabad, Iran

2014 - 2018

B.Sc. in Computer Software Engineering

GPA: A+



AWARDS AND HONORS

- **Research Grant:** Awarded by the University of Udine for the project *"Towards Digital and Smart Agriculture: Use of Artificial Intelligence and Bioinformatic Approaches to Support the Quality of the Agrifood Chain"*
- **M.Sc. Thesis:** Completed with an excellent grade
- **Award Recognition:** Selected as one of the Best Papers at the International Conference on Information and Knowledge Technology (IKT) 2021
- **Ranked Second in GPA:** Among the graduating M.Sc. class at the University of Isfahan
- **Direct M.Sc. Admission:** Awarded in IT Engineering at the University of Isfahan, exempted from the Iranian University Entrance Exam
- **Ranked First in GPA:** Among the graduating B.Sc. class at Lorestan University

PROFESSIONAL EXPERIENCE

University of Udine

2024 - 2025

AI Researcher — Machine Learning and Explainable AI

- Developed ML models for high-dimensional biological and environmental datasets (spectral, microbiome, transcriptomic), leading to three co-authored research papers.
- Applied SHAP to identify key predictive features and improve model interpretability.
- Collaborated with interdisciplinary teams in bioinformatics, plant science, and agrifood research.

University of Isfahan (UI)

2022 - 2024

Research Assistant

- Co-authored in the paper *"Implementing Double-Blind Peer Assessment in Virtual Classroom"*; Contributed to the EU-Iran STEM/UNITEL project 2023, enhancing student engagement and learning outcomes through anonymous feedback systems.
- Co-authored in the paper *"Combinatorial Auction Based on Social Choice in the Internet of Things"*; Investigated social choice mechanisms to optimize resource allocation and decision-making processes in IoT environments.

Fall 2018

Teaching Assistant

- Fundamentals of Electronic Marketplaces
- Artificial Intelligence



SELECTED PUBLICATIONS

- **Fatemeh Hassanvand**, Samuele Boga, Ali Abdari, Slobodanka Radovic, Giuseppe Serra, and Fabio Marroni; Microbiome-Based Classification of Soil Conditions using Machine Learning and Explainable AI; Under review.
- Michele Canciani; **Fatemeh Hassanvand**; Ali Abdari; Arianna Lodovici; Francesco Flagiello; Seyedehfatemeh Kiaeianmoosavi; Giuseppe Serra; Laura Zanin; Iron Deficiency Detection in Plants: Machine Learning to Uncover Significant Wavelengths in Reflectance Spectra; under review.
- Arianna Lodovici, Leilei Zhang, Nicola Tomasi, Gabriella Vinci, Fabio Marroni, Barbara Piani, **Fatemeh Hassanvand**, Mustapha Arkoun, Luigi Lucini, and Laura Zanin; The Early Response to Urea, Nitrate, or Ammonium and Iron Resupply in Tomato Roots Highlights the Induction of FER, bHLHs, UMAMITs and MATEs; Accepted.
- **Fatemeh Hassanvand**, Faria Nassiri-Mofakham, and Katsuhide Fujita; *Automated Negotiation Agents in Modeling Single-Peaked Bidders: An Experimental Comparison*; Information 2024, 15, 508. <https://doi.org/10.3390/info15080508>.
- Faria Nassiri-Mofakham, **Fatemeh Hassanvand**, and Niloofar Mojooodi; *Implementing Double-Blind Peer Assessment in Virtual Classroom*; In 2024 11th International and the 17th National Conference on E-Learning and E-Teaching (ICeLeT) (pp. 1-6); IEEE. [10.1109/ICeLeT62507.2024.10493064](https://doi.org/10.1109/ICeLeT62507.2024.10493064).
- Maedeh Esmaili, Faria Nassiri-Mofakham, **Fatemeh Hassanvand**; (2023, December). *Combinatorial Auction Based on Social Choice in the Internet of Things*; In 2023 14th International Conference on Information and Knowledge Technology (IKT) (pp. 148-152). IEEE. [10.1109/IKT62039.2023.10433061](https://doi.org/10.1109/IKT62039.2023.10433061).
- **Fatemeh Hassanvand**, Faria Nassiri-Mofakham; (2023). *Automated Negotiation Agents in Modeling Gaussian Bidders*; AUT Journal of Modeling and Simulation, 55(1), 3-16. [10.22060/miscj.2023.21153.5273](https://doi.org/10.22060/miscj.2023.21153.5273).
- **Fatemeh Hassanvand**, Faria Nassiri-Mofakham; *Experimental analysis of automated negotiation agents in modeling Gaussian bidders*; In 2021 12th International Conference on Information and Knowledge Technology (IKT) (pp. 197-201). IEEE. [10.1109/IKT54664.2021.9685464](https://doi.org/10.1109/IKT54664.2021.9685464).
- Zahra Karimi, Marjan Kaedi, **Fatemeh Hassanvand**; A Persian Translation of "Using Z: Specification, Refinement, and Proof" by Jim Woodcock & Jim Davies. Access from [here](#).

SKILLS

Languages	English (C1, Duolingo English Test Score: 140/160), Italian (A2), Persian (Native)
Coding	Python, Java, SQL
Databases	MySQL, SQL Server, Redis, MongoDB
ML Frameworks	NumPy, Pandas, Matplotlib, scikit-learn, SHAP

EXTRA COURSES

- PERSEO Winter School "Ethics and Interdisciplinary Research Methods for Personalized Robotics"
- AI for Medical Diagnosis, Coursera.
- Advanced Learning Algorithms, Coursera.
- Neural Networks and Deep Learning, Coursera.
- Supervised Machine Learning: Regression and Classification, Coursera.
- Python for Data Science, AI & Development, Coursera.

REFERENCES

- Dr. Laura Zanin, Researcher at Dep. of Agricultural, Food, Environmental and Animal Sciences, University of Udine, **Email:** laura.zanin@uniud.it
- Dr. Fabio Marroni, Researcher at Dep. of Agricultural, Food, Environmental and Animal Sciences, University of Udine, **Email:** fabio.marroni@uniud.it
- Dr. Faria Nassiri-Mofakham, Faculty of Computer Engineering, University of Isfahan, Email: fnasiri@eng.ui.ac.ir